

Exploring Heat Wave Characteristics and Weekly Mortality Across OECD Regions (UK, Spain, Italy & France)

Research Question: How do heat wave intensity, duration, and time-from-last-heatwave influence mortality across OECD regions?

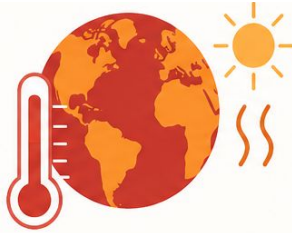


Climatematch
Academy

Group members:

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- 2) Pooja S
- 3) Alon Itzkovitch
- 4) Mark Zakharzhevsky

Research Question: How do heat wave **intensity**, **duration**, and **time-from-the-last-heatwave** influence mortality across OECD regions?



Climate change is increasing heatwaves worldwide

Heatwaves increase
mortality risk



Heatwaves
are not all the same



Knowing what drives heatwave
impact helps build better
warnings and targeted action



Data & Methodology

1) [ERA5](#) dataset (Copernicus Climate):

Variable: 2 m Temperature (t_{2m}),

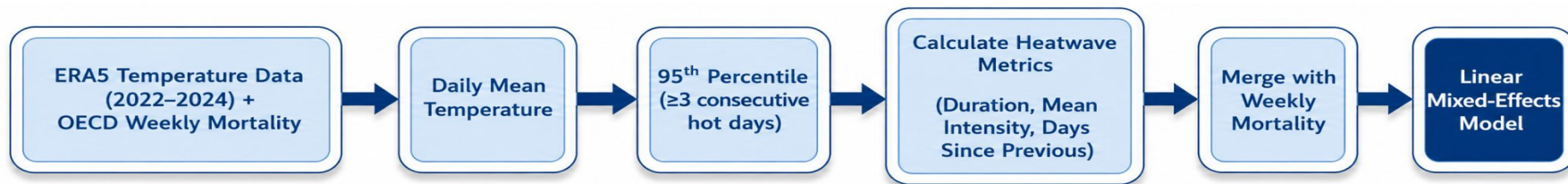
Time resolution: Hourly

2) [Weekly Mortality Data](#) for OECD regions (Eurostat)

Additional data:

[Constrained estimates](#) of 2015-2030 total number of people per grid square

at a resolution of 30 arc (WorldPop - School of Geography and Environmental Science, University of Southampton)



Data preprocessing

Country level aggregation:

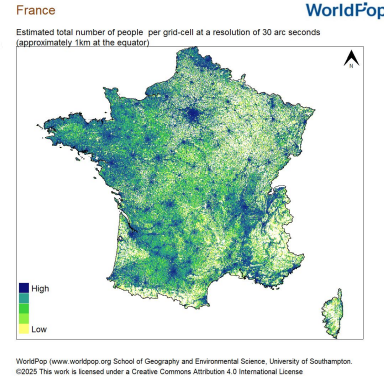
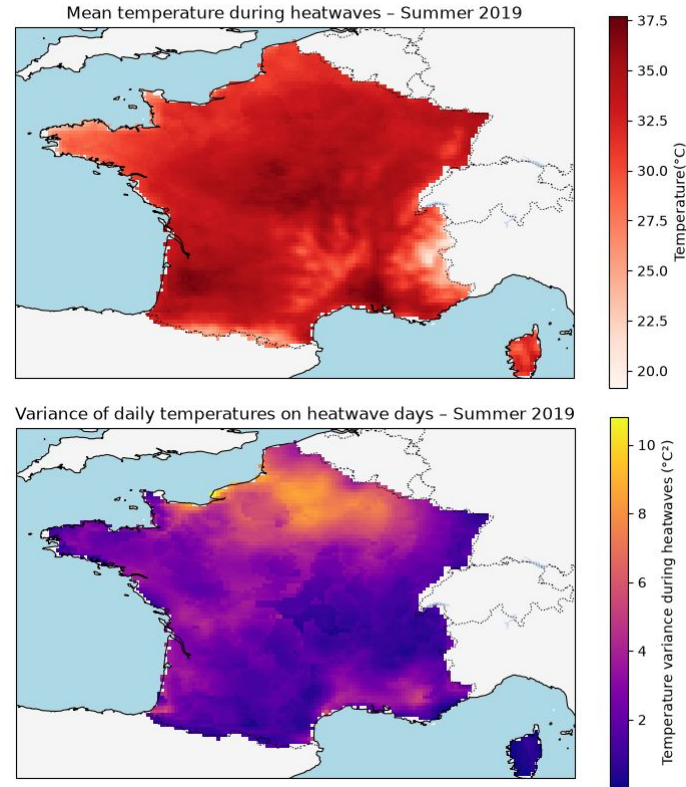
Uniform for Spain, UK and Italy, population weighted for France

Heatwave detection:

Threshold of 95 percentile, period of more than 3 consecutive exceeding days.
Country level data of 2015-2025 for UK, 1990-2024 for France.

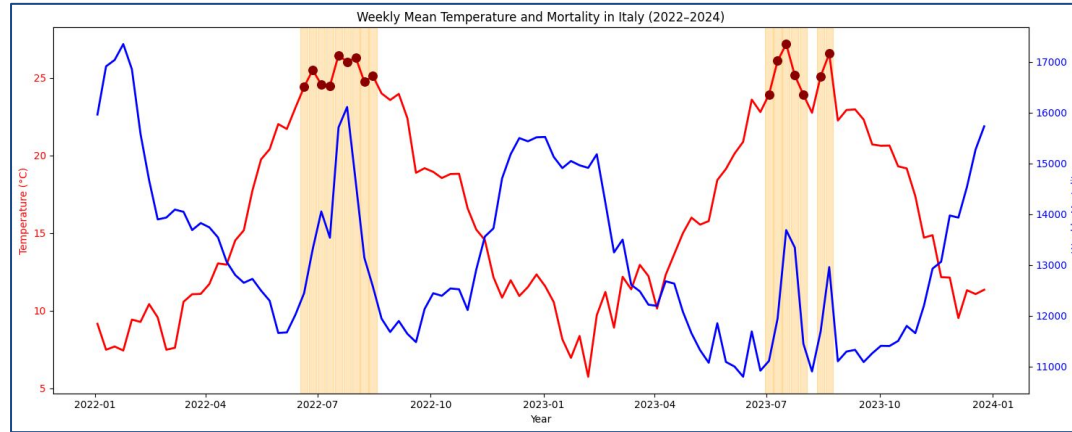
Descriptor space generation:

Calculation of time since last heatwave, duration and etc.



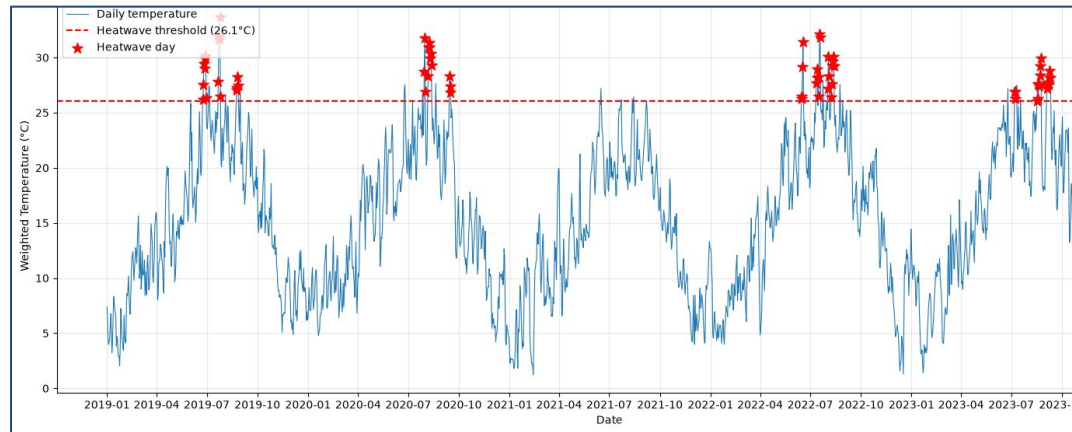
Spatial characteristics of temperature and population.

Heatwave Occurrence and Mortality



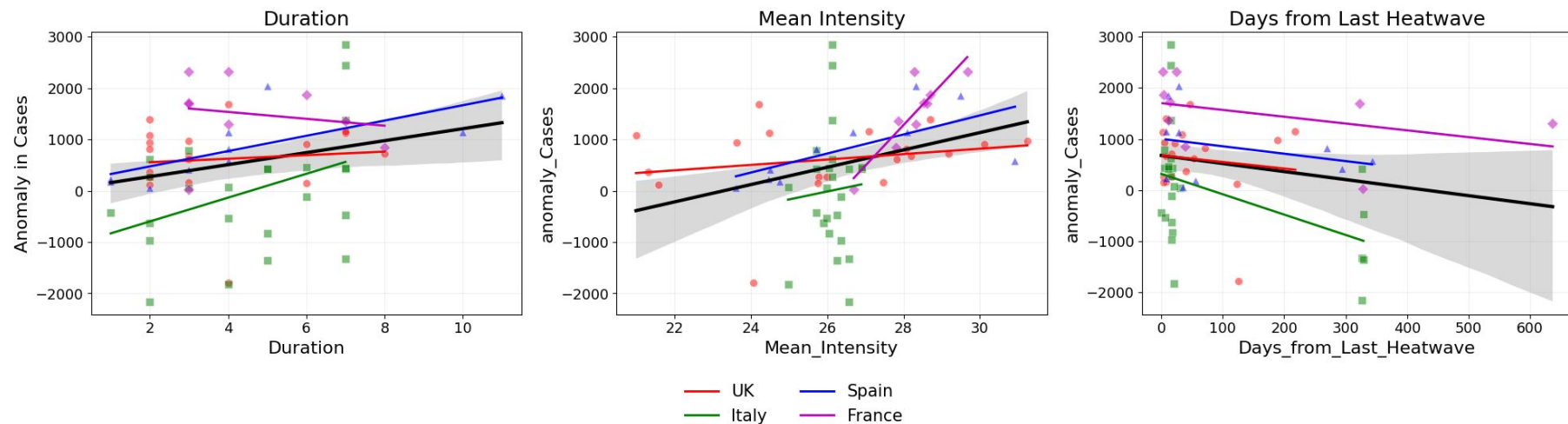
Heatwave thresholds, °C:

- France: 26.1
- Spain: 22.9
- Italy: 25.1
- UK: 21.6



How heatwaves characteristics affect excess mortality cases?

Excess Mortality Cases ~ Duration + Mean Intensity + Days from Last Heatwave + (1|Country)



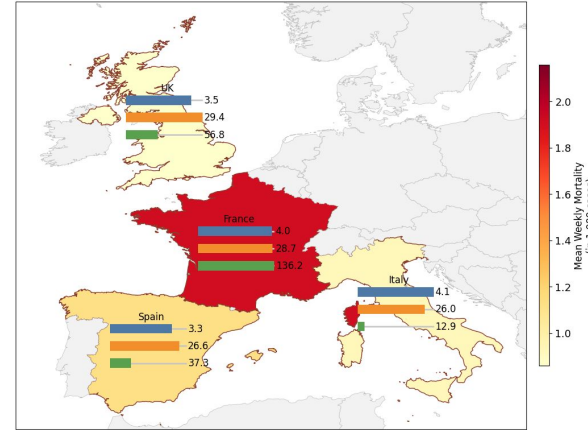
Longer heatwaves and shorter interval (in days) since the previous heatwave are associated with an increase in mortality (compared to expected mortality).

Conclusion

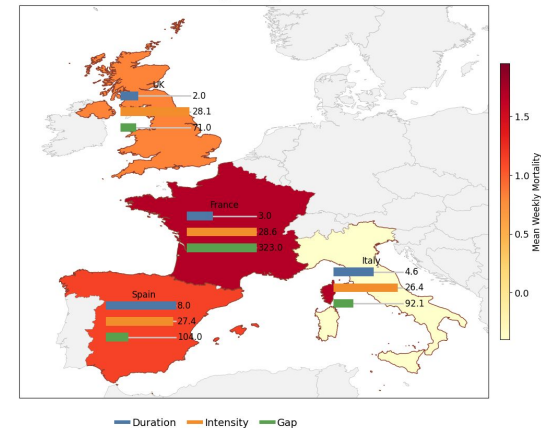
1. Heatwaves characteristics are linked to excess mortality
2. Heatwaves effects **are not uniform** across regions or over the years.
3. Regional differences indicate that **local climate, adaptation, population vulnerability, and other factors** can modify the health impacts of heatwaves, highlighting the importance of region-specific risk assessment and intervention strategies.

Heatwave Characteristics and Mortality
Anomaly from mean number of Cases

Summer [Jun-Aug], 2022

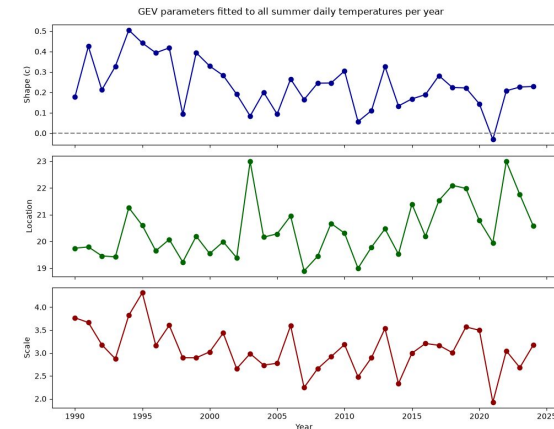
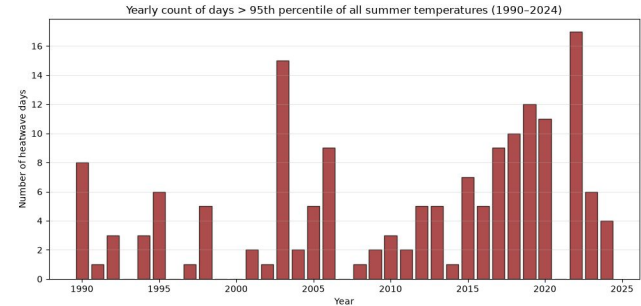


Duration Intensity Gap
Summer [Jun-Aug], 2024

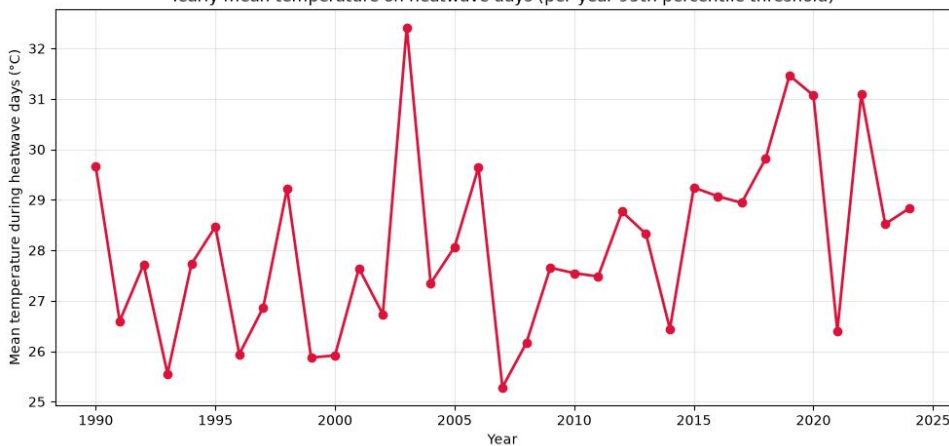


Future directions

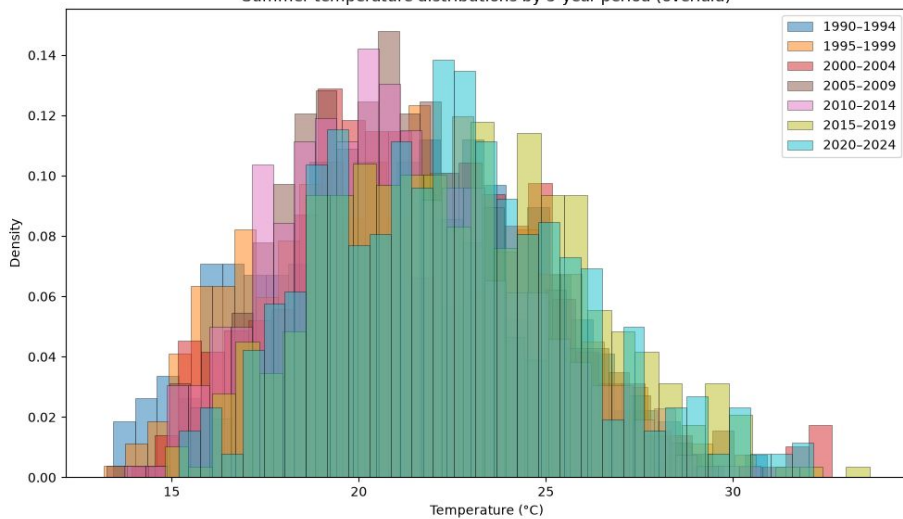
- Country-level analysis -> City level analysis
- 2022-2024 research period -> Increased research period:
- Temperature based variables -> Climate, demographic and environmental variables
- Addition of dynamics of temperature change
- Weekly mortality data for a heatwave- > daily mortality data with bigger window surrounding the heatwave
- All-cause mortality - > heat associated mortality causes
- Climate change and variability analysis



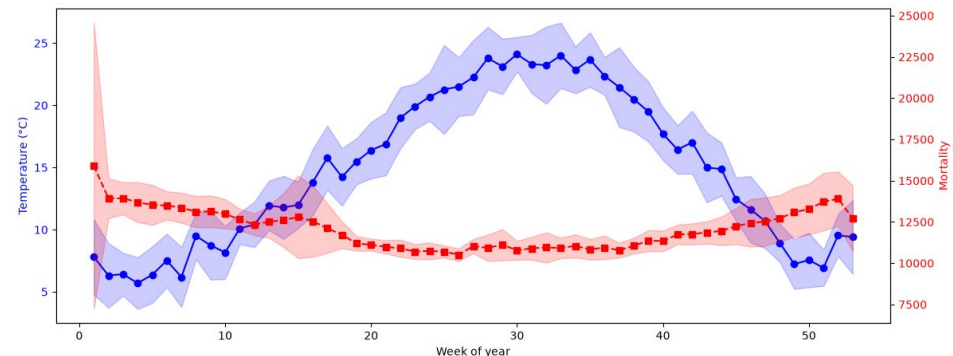
Yearly mean temperature on heatwave days (per-year 95th percentile threshold)



Summer temperature distributions by 5-year period (overlaid)

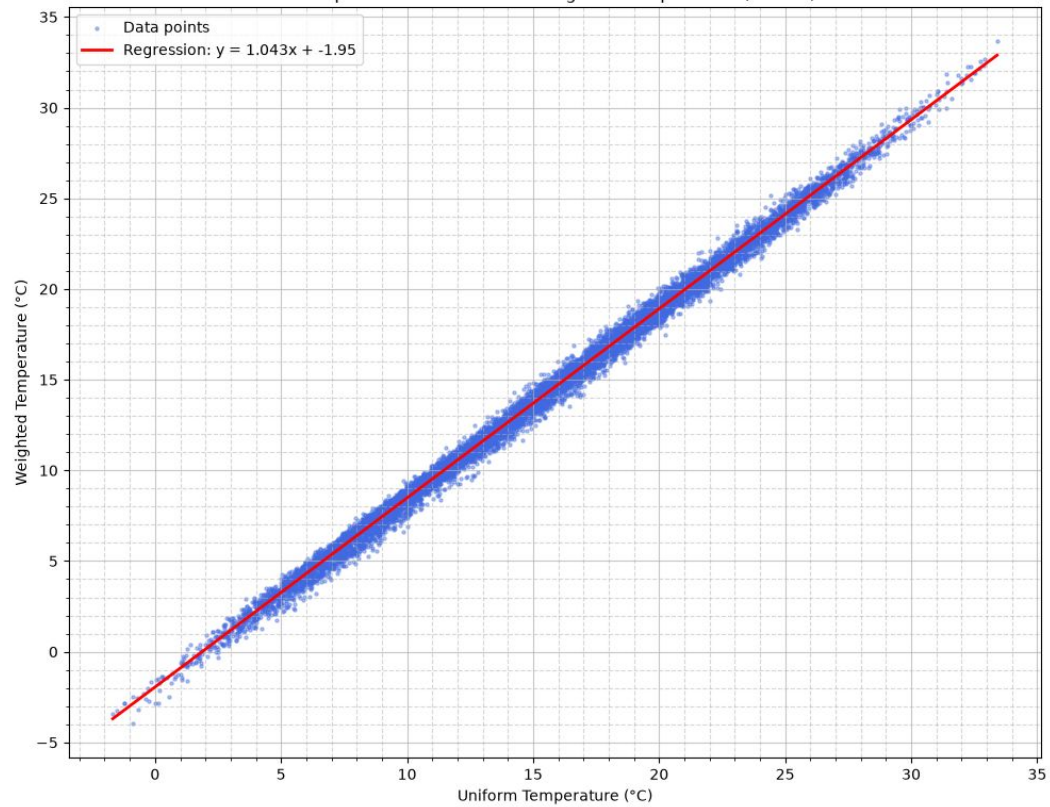


Mean weekly temperature and mortality with 1 STD shading, 2015 - 2023



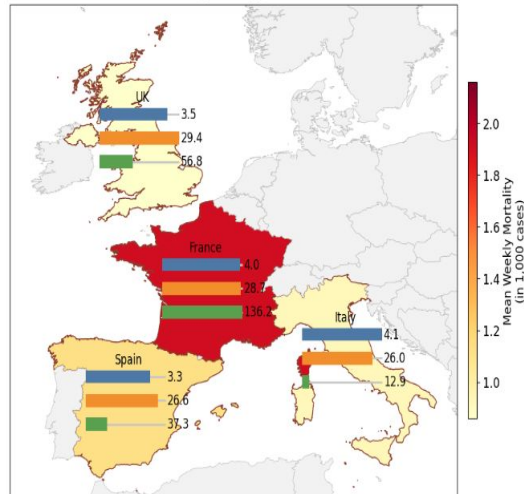
France climate variability

Comparison of Uniform vs Weighted Temperature (France)



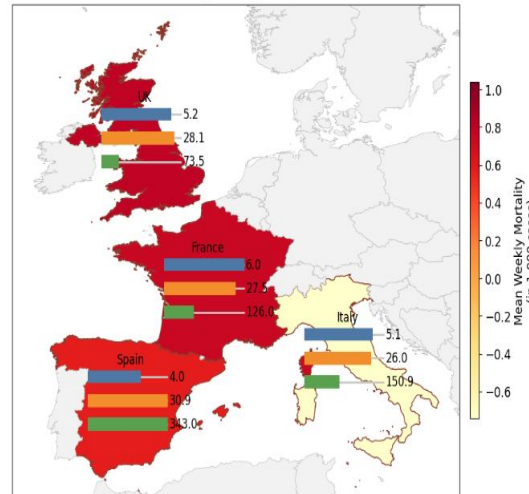
Heatwave Characteristics and Mortality Anomaly from mean number of Cases

Summer [Jun-Aug], 2022



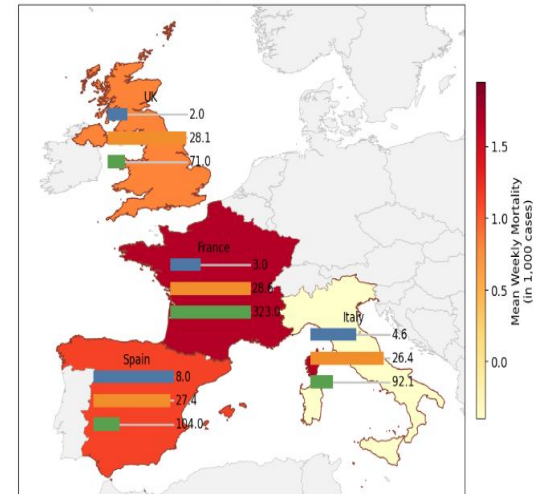
Duration Intensity Gap

Summer [Jun-Aug], 2023



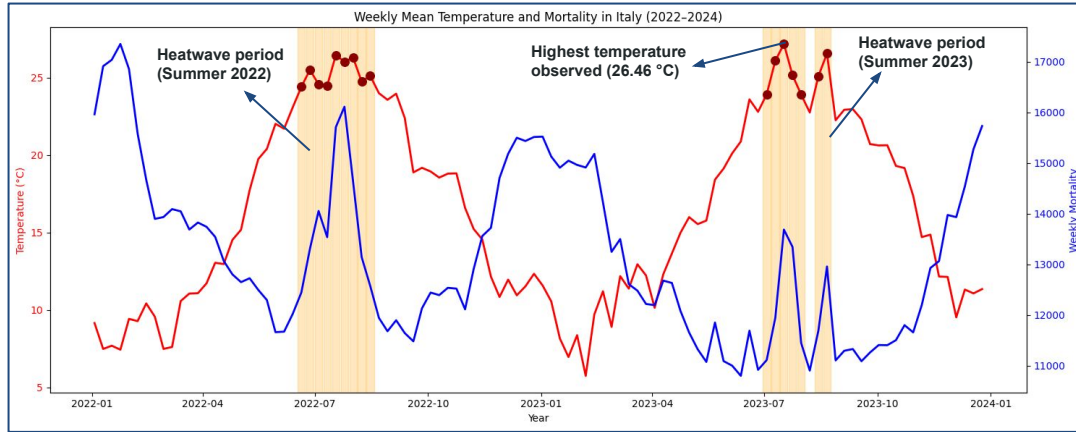
Duration Intensity Gap

Summer [Jun-Aug], 2024



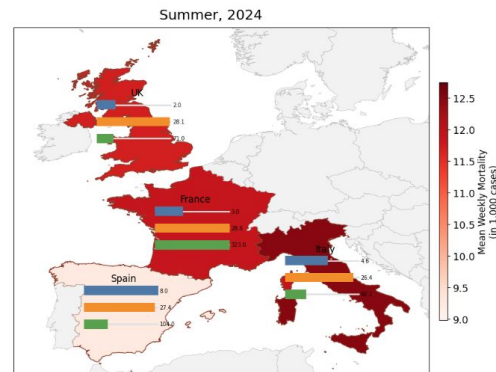
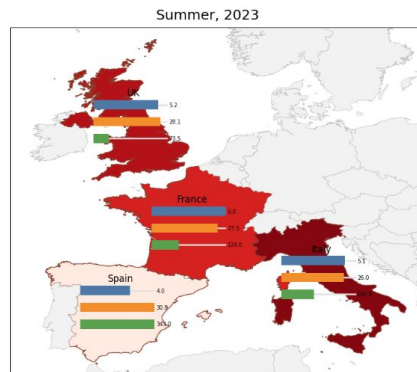
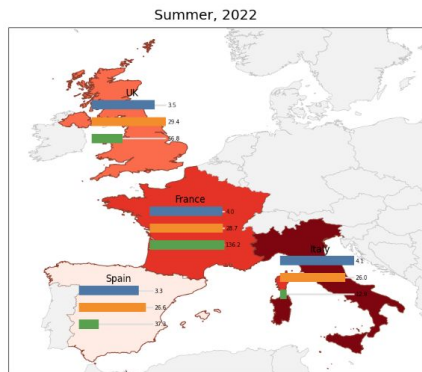
Duration Intensity Gap

Mortality During Heatwaves

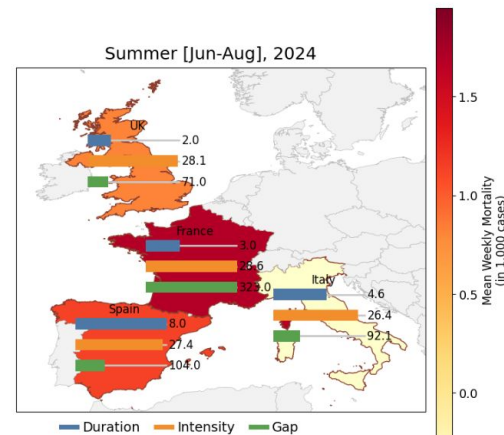
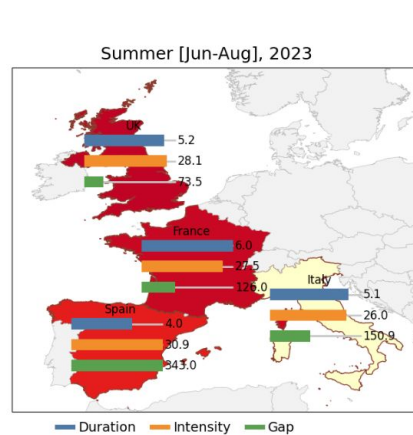
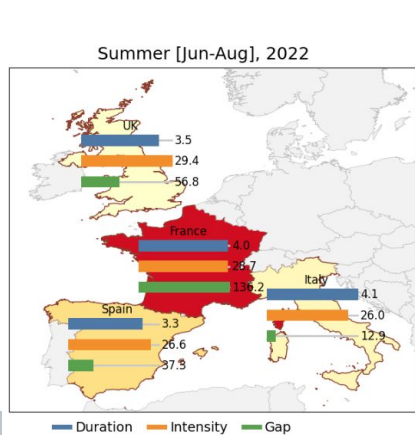


Weekly Mean Temperature and Mortality in Italy (2022-2024)

Heatwave Characteristics and Mortality Raw number of Cases



Heatwave Characteristics and Mortality Anomaly from mean number of Cases



Mixed Linear Model Regression Results

Model:	MixedLM	Dependent Variable:	Cases
No. Observations:	62	Method:	REML
No. Groups:	4	Scale:	1047454.7097
Min. group size:	9	Log-Likelihood:	-504.4728
Max. group size:	24	Converged:	Yes
Mean group size:	15.5		

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	13734.822	2078.063	6.609	0.000	9661.893	17807.750
Duration	174.027	61.706	2.820	0.005	53.086	294.967
Mean_Intensity	-105.753	72.650	-1.456	0.145	-248.144	36.639
Days_from_Last_Heatwave	-1.722	1.006	-1.712	0.087	-3.693	0.249
Group Var	3245943.832	2735.151				

Mixed Linear Model Regression Results

Model:	MixedLM	Dependent Variable:	anomaly_Cases
No. Observations:	62	Method:	REML
No. Groups:	4	Scale:	707916.2893
Min. group size:	9	Log-Likelihood:	-490.4524
Max. group size:	24	Converged:	Yes
Mean group size:	15.5		

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	-1847.538	1583.981	-1.166	0.243	-4952.084	1257.009
Duration	98.549	50.748	1.942	0.052	-0.916	198.014
Mean_Intensity	87.964	60.346	1.458	0.145	-30.311	206.240
Days_from_Last_Heatwave	-2.270	0.827	-2.744	0.006	-3.892	-0.649
Group Var	324370.388	384.814				

Future Work

Improve spatial resolution

- Analyze regional, city-level, and **urban vs. rural** differences in heatwave impacts.

Expand the dataset

- Include additional European countries and extend the study period beyond 2022–2024.

Incorporate additional variables

- Account for humidity, air pollution, socioeconomic factors, age distribution, and healthcare accessibility.

Explore advanced modeling

- Investigate lag effects, non-linear relationships, and machine learning approaches to better predict heat-related mortality.

Support climate adaptation

- Develop evidence to inform heat-health warning systems and targeted public health interventions.